# BUSS386 Problem Set 8

Trading Strategies Involving Options

Prof. Ji-Woong Chung

## Problem 1 — Synthetic positions and a parity check

A European call on a non-dividend stock with $K = 50$ and $T = 0.5$ years trades at \$4.20. $S_0 = 51$ and $r = 4\%$ (c.c.).

(a) What price does put-call parity imply for the European put with the same strike and maturity?

(b) The put actually trades at \$2.80. Describe the arbitrage: list the four legs, the cash flow today, and show that the position pays zero at $T$ whatever $S_T$ is.

(c) You want a short stock position but cannot borrow the shares. Build a synthetic short from the call, the put, and a bond with face value $K$ at $T$. List the three legs, the cash received today at market prices, and the payoff at $T$.

(d) In one sentence: when would a trader prefer the synthetic short to an actual short sale?

## Problem 2 — Principal-protected note (ELB)

A securities firm sells a 1-year equity-linked bond (ELB) on the KOSPI 200. You invest ₩ 10,000,000; $r = 3\%$ (c.c.). A 1-year at-the-money European call on ₩ 10,000,000 of index costs ₩ 850,000. Assume the firm charges no fee.

(a) How much does the firm spend on a zero-coupon bond that returns the principal at maturity, and how much is left over?

(b) The leftover buys a fraction of one call. What is the *participation rate* of the note?

(c) What does the note pay at maturity if the index ends 20% higher? 20% lower?

(d) Repeat (a)–(b) with $r = 1\%$. In one sentence, why do ELB participation rates rise and fall with deposit rates?

(e) Using put-call parity, describe a different portfolio of index, puts and bonds that has the same payoff as the note.

## Problem 3 — The ELS as bond plus short put

A 3-year ELS linked to an index pays an 8% annual coupon (simple, paid at maturity) and the principal, provided the index at maturity is at or above 65% of its initial level. If the index is below 65%, the investor receives principal multiplied by the index ratio $I_T/I_0$ and no coupon. Work per ₩ 100 invested, and treat the barrier as observed only at maturity.

(a) Tabulate the payoff for $I_T/I_0 \in \{0.40,\ 0.60,\ 0.65,\ 0.80,\ 1.20\}$.

(b) Deposits pay 3%. The ELS pays 8%. What has the investor sold to earn the extra 5% a year?

(c) Write the payoff as a bond plus option positions on the index ratio. (Hint: compare it with "bond paying 124 minus a put struck at 0.65" and account for what is left.)

(d) In one sentence, why does the at-maturity picture understate the risk of the actual product?

## Problem 4 — Covered call on a KOSPI 200 ETF

KODEX KOSPI 200 ETF trades at ₩ 40,000 per share. A 1-month call on the ETF with $K =$ ₩ 41,000 trades at ₩ 500. You own 1,000 shares and write 1,000 covered calls. Ignore the time value of money and contract-size adjustments.

(a) Compute the total premium received.

(b) Compute total portfolio value at expiry for $S_T \in \{38{,}000;\ 40{,}000;\ 41{,}000;\ 42{,}000;\ 45{,}000\}$.

(c) What is the maximum portfolio value, and at what $S_T$ is it reached?

(d) What is the breakeven $S_T$ relative to the original ₩ 40,000,000?

(e) Using put-call parity, express the covered call as a bond plus a single option. Check the equivalence at $S_T = 38{,}000$.

(f) A "monthly dividend" covered-call ETF runs this position every month. If the index jumps to ₩ 48,000 within the month, by how much does the fund lag a plain index ETF?

## Problem 5 — Collar, or forward?

A founder owns 1,000,000 shares of her company's stock at \$80 and wants protection for one year; $r = 4\%$ (c.c.). Available:

- 1-year put with $K_p =$ \$70 at \$3.50 per share.
- 1-year call with $K_c =$ \$95 at \$3.50 per share.
- 1-year forward on the stock at the no-arbitrage forward price.

(a) Verify that the collar (long put, short call) is zero-cost.

(b) Compute the collared position's value at expiry for $S_T \in \{60, 70, 80, 95, 110\}$ per share, in total dollars.

(c) What are the minimum and maximum values of the collared position?

(d) Compute the forward price and the value at expiry of the position hedged with a short forward, for the same five values of $S_T$.

(e) In one sentence each: why might the founder prefer the collar to selling the stock, and the collar to the forward?

## Problem 6 — Bull and bear spreads: the mirror rule

A stock trades at \$50. One-year European options are quoted ($r = 0$ for this problem):

| | $K = 45$ | $K = 55$ |
|---|---|---|
| Call | \$8 | \$3 |
| Put | \$3 | \$8 |

(a) Build a bull spread with calls. Net cash today, maximum gain, maximum loss, breakeven.

(b) Build a bull spread with puts. Net cash today, maximum gain, maximum loss, breakeven. Compare with (a).

(c) Build a bear spread with calls and complete the profit table for $S_T \in \{40, 45, 50, 55, 60\}$.

(d) State the mirror rule in one sentence.

(e) The call and put versions of each spread have the same profit diagram. In one sentence, what differs between them in practice?

## Problem 7 — Butterfly and the convexity check

European calls on the same underlying and maturity: $K = 90$ at \$13, $K = 100$ at \$7, $K = 110$ at \$3.

(a) Construct a long butterfly with these three strikes. Net premium today?

(b) Tabulate the payoff at expiry for $S_T \in \{85, 95, 100, 105, 115\}$.

(c) A long butterfly must cost something. Why? Does it here?

(d) The next day the $K = 100$ call is quoted at \$8.50 while the other two are unchanged. Is there an arbitrage? Describe it.

(e) A trader with the same "nothing will happen" view simply sells two $K = 100$ calls at \$7 each. Compare the maximum loss of that position with the butterfly's.

## Problem 8 — Calendar spread

A stock trades at \$100 and $r = 3\%$ (c.c.). A 6-month European call with $K = 100$ trades at \$7.76 and a 1-year European call with $K = 100$ trades at \$11.35. An investor sells the 6-month call and buys the 1-year call.

(a) Net cash flow today?

(b) Six months later the stock is at \$130. The investor settles the short call and sells the long call. Using the Lecture 7 bounds on a European call price, find the smallest and largest possible profit on the whole trade (ignore interest on the initial premium).

(c) Repeat (b) with the stock at \$100 after six months.

(d) Suppose that in case (c) the market prices a 6-month at-the-money call exactly as it did today. What is the profit?

(e) In one sentence, what does a calendar spread bet on that a butterfly does not?

## Problem 9 — Straddle vs. strangle

A trader expects a large move in a stock currently at \$100 over the next month, but is unsure of the direction. Available 1-month options:

- ATM straddle: $K = \$100$ call at \$4 + $K = \$100$ put at \$4.
- OTM strangle: $K_c = \$105$ call at \$2 + $K_p = \$95$ put at \$2.

(a) Compute the cost of each strategy.

(b) For each strategy, find the breakeven values of $S_T$.

(c) For each strategy, compute the profit at $S_T \in \{85, 92, 100, 108, 115\}$.

(d) Which strategy is better if the realized move is small (within $\pm 3$)? If large (beyond $\pm 10$)? One sentence each.

## Problem 10 — Iron condor on SPX 0DTE

SPX is at 5,800. A trader sells an iron condor expiring *today* (0DTE) with strikes 5,750 / 5,780 / 5,820 / 5,850. Net credit received \$8 (index points). Multiplier \$100 per point.

(a) List the four legs (long/short, put/call, strike). Which two of today's spreads is the position made of?

(b) Compute the maximum profit and maximum loss per contract, and the values of $S_T$ at which each occurs.

(c) The trader makes \$800 on most days and loses up to \$2,200 on a few. What must be true about the market's implied daily move versus the realized move for the strategy to be profitable on average?

## Problem 11 — Box spread financing

You execute a long box spread on SPX with $K_1 = 5{,}700$ and $K_2 = 5{,}900$, expiry 6 months out. The four legs cost a net premium of \$197.50 today (per box). $r = 4\%$ c.c. for the same maturity.

(a) Confirm that the box pays a constant $K_2 - K_1 = \$200$ at expiry, regardless of $S_T$.

(b) Compute the interest rate implied by the box price.

(c) Compare it with the 4% market rate: is the box rich or cheap, and what would an arbitrageur do?

(d) In one sentence, why might the box rate differ from the Treasury rate?

## Problem 12 — Matching the strategy to the view

For each situation, name the strategy from this lecture that fits, state whether you pay or receive premium up front, and name the view it expresses (direction, and big move or calm).

(i) Mildly bullish on a stock; a plain call looks too expensive.

(ii) You hold a large position, want a floor, and refuse to pay any premium.

(iii) Earnings are released tomorrow; you expect a large move but have no idea which way.

(iv) You believe the stock will be almost exactly where it is now in three months, and you want a known worst case.

(v) You want to lend cash for six months at a rate above the deposit rate.

(vi) You are bearish on a KOSPI 200 stock, and short-selling has just been banned.

(vii) You own shares, expect a dull quarter, and want monthly income.